

MedAssist SLM

Challenge 3 — Impact Statement

Purpose of the Project

MedAssist SLM is a domain-specific small language model project designed to provide trustworthy, evidence-based medical information. It addresses a community problem in which people increasingly rely on online search and AI systems for health information, while general-purpose systems can provide inaccurate, outdated, or unsupported answers. This is particularly important for people in underserved and low-resource communities where access to healthcare professionals and reliable health information may be limited. The project aims to provide accessible health education while clearly communicating the limits of AI and directing users toward professional care when questions require clinical judgment.

Positive Impacts

The most significant potential impact is improved access to reliable health information for individuals, families, caregivers, and communities that may face barriers to timely healthcare information. By grounding responses in reputable medical sources and making those sources traceable, MedAssist could help users better understand common health topics, recognise when professional attention may be necessary, and make more informed decisions about seeking care. At a broader level, the project could demonstrate how smaller, resource-conscious AI systems can be developed responsibly for high-impact domains and adapted to the needs of communities that are often less represented in mainstream AI development.

Potential Risks and Harms

The most significant risk is that users could receive or act on incorrect medical information, particularly if they interpret an AI response as a diagnosis or personalised treatment recommendation. Other risks include bias in the underlying medical literature, underrepresentation of low-resource populations, privacy concerns, overreliance on the system, and the possibility that outdated clinical guidance could be presented as current. There may also be resource and environmental costs associated with training and operating AI systems.

Values and Mitigation

The project is guided primarily by scientific rationalism, universalism, and liberty, alongside safety, privacy, transparency, fairness, and accountability. Scientific rationalism motivates the use of credible medical evidence; universalism informs our goal of improving access across different communities; and liberty supports users' ability to make informed decisions. To reduce risks, the team will prioritise authoritative sources, document data provenance and limitations, avoid identifiable patient information, evaluate the system on held-out questions, implement refusal and escalation mechanisms for high-risk requests, and use disclaimers to distinguish general information from professional medical advice. High-risk content will receive additional review where feasible.

Limitations and Uncertainties

MedAssist will not represent all medical conditions, populations, languages, or healthcare systems equally. The available evidence may be dominated by English-language research and populations from higher-income settings, creating representation gaps. The team also cannot guarantee that generated responses will always be correct. The effectiveness of the system will therefore depend on source quality, retrieval performance, evaluation quality, and how users interpret its responses. These uncertainties will be documented rather than hidden, and findings from evaluation and testing will be

used to refine the system.

Team Reflection

Completing the Problem Statement, Data Card, and Impact Statement changed our view of responsible AI from something added after development into something that must shape the project from the beginning. The Problem Statement made us focus on the people and community need rather than the technology. The Data Card made us question where our medical knowledge comes from, who may be missing from it, and how decisions about collection and labeling can introduce bias. The Impact Statement further encouraged us to consider consequences beyond model accuracy, including misuse, privacy, overreliance, representation, and environmental cost. Together, these activities have made us more deliberate about transparency, safety, inclusion, and accountability. We now see responsible AI as an ongoing process of making assumptions visible, testing for unintended harms, listening to affected communities, and revising the system as new evidence and limitations emerge.